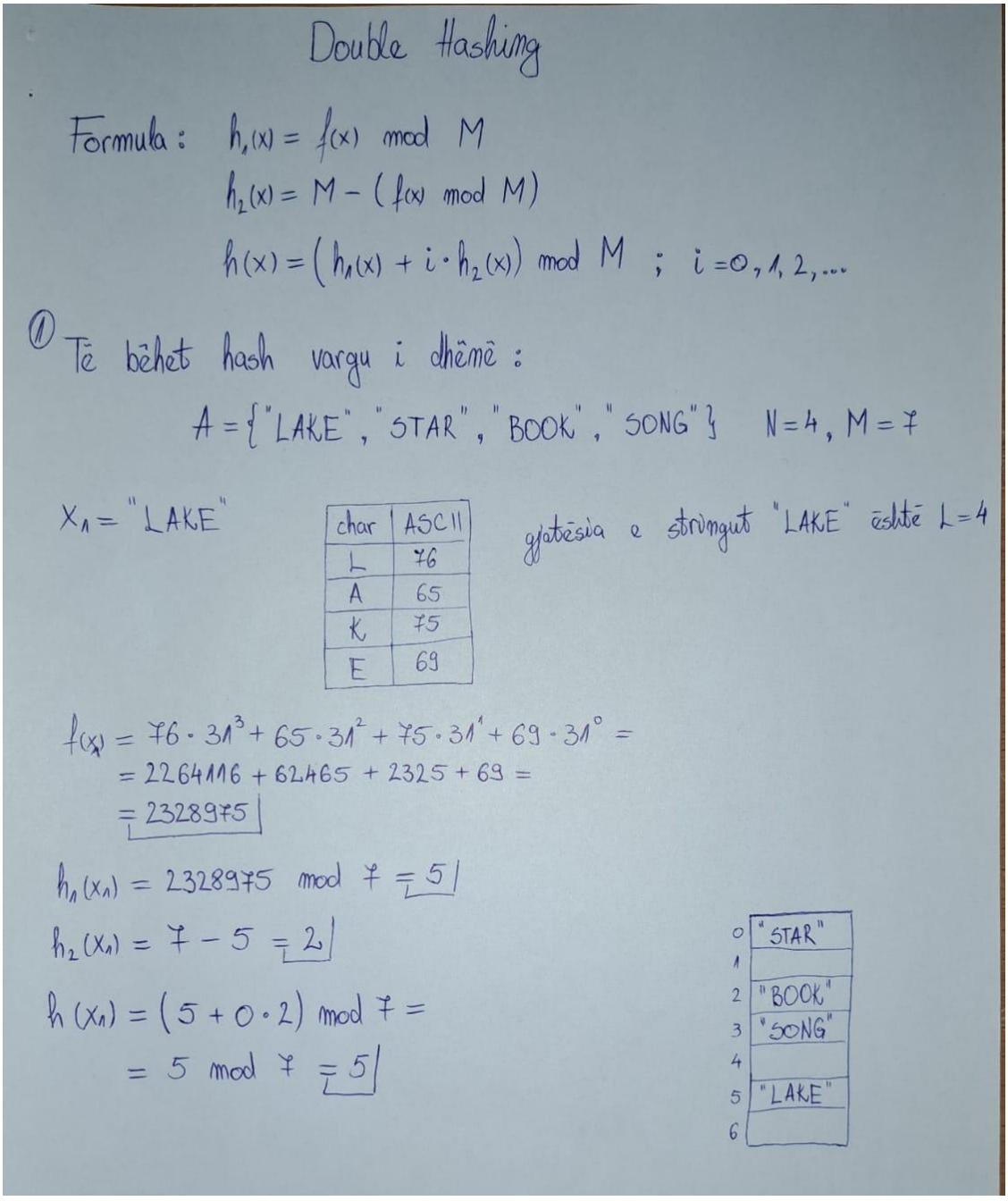
Java 12 - Double Hashing with Strings & Cuckoo Hashing - Detyra të zgjidhura

Lënda: Hyrje në Algoritme

Double Hashing



$X_2 =$ "STAR"

char	ASCII
S	83
T	84
A	65
R	82

stringu "STAR" ka gjatësi $L = 4$

$f(x_2) = 83 \cdot 31^3 + 84 \cdot 31^2 + 65 \cdot 31^1 + 82 \cdot 31^0 =$

$= 2472653 + 80724 + 2015 + 82 = 2555474$

$h_1(x_2) = 2555474 \mod 7 = 5$ collision

$h_2(x_2) = 7 - 5 = 2$

$h(x_2) = (5 + 1 \cdot 2) \mod 7 =$

$= 7 \mod 7 = 0$

$X_3 =$ "BOOK"

char	ASCII
B	66
O	79
O	79
K	75

stringu "BOOK" ka gjatësi $L = 4$

$f(x_3) = 66 \cdot 31^3 + 79 \cdot 31^2 + 79 \cdot 31^1 + 75 \cdot 31^0 =$

$= 1966206 + 75919 + 2449 + 75 = 2044649$

$h_1(x_3) = 2044649 \mod 7 = 5$ / collision

$h_2(x_3) = 7 - 5 = 2$

$h(x_3) = (5 + 1 \cdot 2) \mod 7 =$

$= 7 \mod 7 = 0$ collision

$h(x_3) = (5 + 2 \cdot 2) \mod 7 =$

$= 9 \mod 7 = 2$

$X_4 =$ "SONG"

char	ASCII
S	83
O	79
N	78
G	71

stringu "SONG" ka garumu $L = 4$

$$f(X_4) = 83 \cdot 31^3 + 79 \cdot 31^2 + 78 \cdot 31^1 + 71 \cdot 31^0 =$$
$$= 2472653 + 75919 + 2418 + 71 = \boxed{2551061}$$

$h_1(X_4) = 2551061 \mod 7 = \boxed{2}$ collision

$h_2(X_4) = 7 - 2 = \boxed{5}$

$h(X_4) = (2 + 1 \cdot 5) \mod 7 =$
$= 7 \mod 7 = \boxed{0}$ collision

$h(X_4) = (2 + 2 \cdot 5) \mod 7 =$
$= 12 \mod 7 = \boxed{5}$ collision

$h(X_4) = (2 + 3 \cdot 5) \mod 7 =$
$= 17 \mod 7 = \boxed{3}$

Detyra 2: Të bëhet hash vargu i dhënë

$A = \{$ "bean", "date", "kiwi", "corn" $\}$ me Double Hashing

→ $N = 4$, $M = 7$

Formulat: $h(x) = (h_1(x) + i \cdot h_2(x)) \bmod M$

$h_1(x) = f(x) \bmod M$

$h_2(x) = M - (f(x) \bmod M)$

$f(x)$ → do të jetë stringu i konvertuar në integer value.

⇒ Strings do të konvertohen në integer me metodën e Horner-it

X_1 = "bean"

char	ASCII
b	98
e	101
a	97
n	110

$L = 4$ → length i string-ut "bean"

$$f(x_1) = 98 \cdot 31^{L-1} + 101 \cdot 31^{L-2} + 97 \cdot 31^{L-3} + 110 \cdot 31^{L-4}$$
$$= 98 \cdot 31^3 + 101 \cdot 31^2 + 97 \cdot 31^1 + 110 \cdot 31^0$$
$$= 98 \cdot 29791 + 101 \cdot 961 + 97 \cdot 31 + 110 \cdot 1$$
$$= 2919518 + 97061 + 3007 + 110 = \boxed{3019696}$$

$h_1(x_1) = (3019696) \bmod 7 = \boxed{1}$

$h_2(x_1) = 7 - (3019696 \bmod 7) = 7 - 1 = \boxed{6}$

$h(x_1) = (1 + 0 \cdot 6) \bmod 7 = 1 \bmod 7 = \boxed{1}$ ✓

0	"corn"
1	"bean"
2	
3	
4	"date"
5	"kiwi"
6	

X_2 = date

char	ASCII
d	100
a	97
t	116
e	101

$L = 4$ → length i string-ut "date"

$$f(x_2) = 100 \cdot 31^{L-1} + 97 \cdot 31^{L-2} + 116 \cdot 31^{L-3} + 101 \cdot 31^{L-4}$$
$$= 100 \cdot 31^3 + 97 \cdot 31^2 + 116 \cdot 31^1 + 101 \cdot 31^0$$
$$= 100 \cdot 29791 + 97 \cdot 961 + 116 \cdot 31 + 101 \cdot 1$$
$$= 2979100 + 93217 + 3596 + 101$$
$$= \boxed{3076014}$$

$h_1(x_2) = (3076014) \bmod 7 = \boxed{4}$ ✓

x_3 = "kiwi"

char	ASCII
k	107
i	105
w	119
i	105

L=4 ⇒ length i string-ut "kiwi"

$f(x_3) = 107 \cdot 31^3 + 105 \cdot 31^2 + 119 \cdot 31^1 + 105 \cdot 31^0$

$= 107 \cdot 29791 + 105 \cdot 961 + 119 \cdot 31 + 105 \cdot 1$

$= 3187637 + 100905 + 3689 + 105$

$= \boxed{3292336}$

$h_1(x_3) = (3292336) \bmod 7 = \boxed{5}$ ✓

x_4 = "corn"

char	ASCII
c	99
o	111
r	114
n	110

L=4 ⇒ length i string-ut "corn"

$f(x_4) = 99 \cdot 31^3 + 111 \cdot 31^2 + 114 \cdot 31^1 + 110 \cdot 31^0$

$= 99 \cdot 29791 + 111 \cdot 961 + 114 \cdot 31 + 110 \cdot 1$

$= 2949309 + 106671 + 3534 + 110$

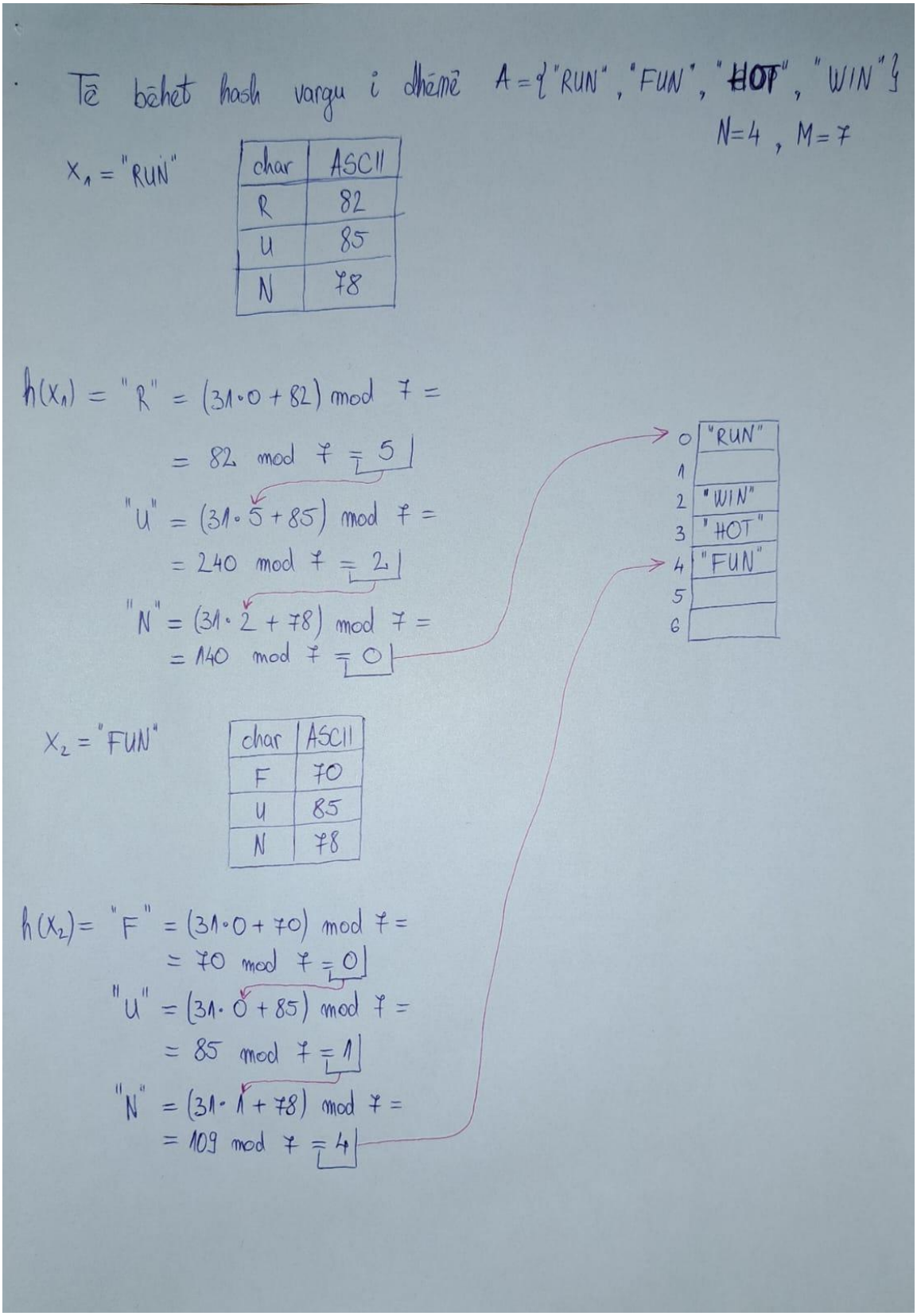
$= \boxed{3059624}$

$h_1(x_4) = (3059624) \bmod 7 = \underline{1}$, collision, i++

$h_2(x_4) = 7 - (3059624 \bmod 7) = 7 - 1 = \underline{6}$

$h(x_4) = (1 + \underset{\text{(i)}}{1} \cdot 6) \bmod 7 = 7 \bmod 7 = \boxed{0}$ ✓

Rolling Hash



$X_3 =$ "HOT"

char	ASCII
H	72
O	79
T	84

$h(X_3) =$ "H" $= (31 \cdot 0 + 72) \bmod 7 =$
$= 72 \bmod 7 = \boxed{2}$

"O" $= (31 \cdot 2 + 79) \bmod 7 =$
$= 141 \bmod 7 = \boxed{1}$

"T" $= (31 \cdot 1 + 84) \bmod 7 =$
$= 115 \bmod 7 = \boxed{3}$

$X_4 =$ "WIN"

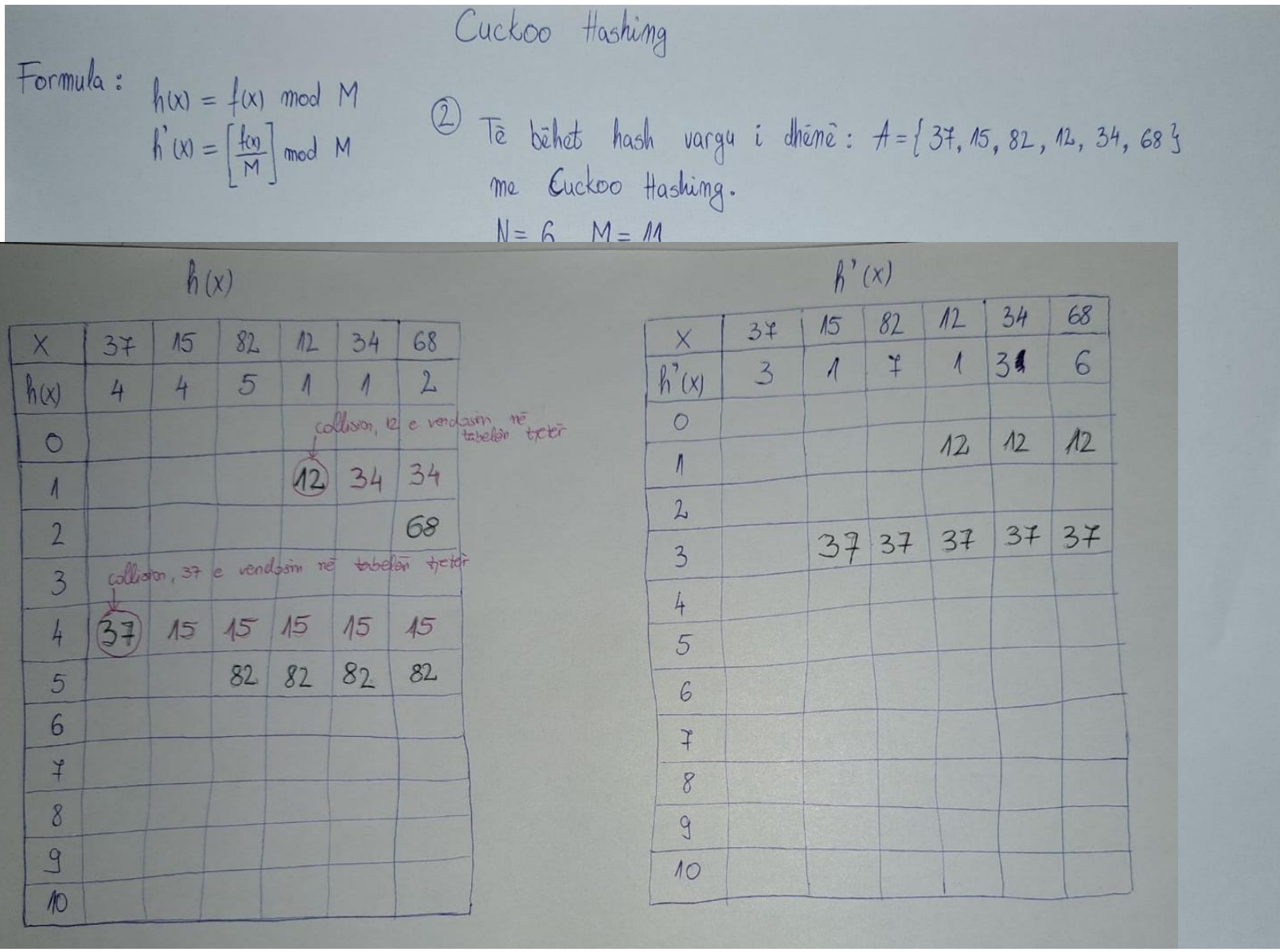
char	ASCII
W	87
I	73
N	78

$h(X_4) =$ "W" $= (31 \cdot 0 + 87) \bmod 7 =$
$= 87 \bmod 7 = \boxed{3}$

"I" $= (31 \cdot 3 + 73) \bmod 7 =$
$= 166 \bmod 7 = \boxed{5}$

"N" $= (31 \cdot 5 + 78) \bmod 7 =$
$= 233 \bmod 7 = \boxed{2}$

Cuckoo Hashing



Cuckoo Hashing

③ Të bëhet hash vargu i dhënë $A = \{"UBT", "CSE", "TIK", "TOK"\}$ me Cuckoo Hashing $N = 4$, $M = 7$

$X_1 = "UBT"$

char	ASCII
U	85
B	66
T	84

$f(x_1) = 85 \cdot 31^2 + 66 \cdot 31^1 + 84 \cdot 31^0 = 81685 + 2046 + 84 = 83815$

$X_2 = "CSE"$

C	67
S	83
E	69

$f(x_2) = 67 \cdot 31^2 + 83 \cdot 31^1 + 69 \cdot 31^0 = 64387 + 2573 + 69 = 67029$

$X_3 = "TIK"$

T	84
I	73
K	75

$f(x_3) = 84 \cdot 31^2 + 73 \cdot 31^1 + 75 \cdot 31^0 = 80724 + 2263 + 75 = 83062$

$X_4 = "TOK"$

T	84
O	79
K	75

$f(x_4) = 84 \cdot 31^2 + 79 \cdot 31^1 + 75 \cdot 31^0 = 80724 + 2449 + 75 = 83248$

$h(x_1) = 83815 \bmod 7 = 4$

$h(x_2) = 67029 \bmod 7 = 4$

$h(x_3) = 83062 \bmod 7 = 0$

$h(x_4) = 83248 \bmod 7 = 4$

$h'(x_1) = \left(\frac{83815}{7}\right) \bmod 7 = 3$

$h'(x_2) = \left(\frac{67029}{7}\right) \bmod 7 = 6$

$h'(x_3) = \left(\frac{83062}{7}\right) \bmod 7 = 1$

$h'(x_4) = \left(\frac{83248}{7}\right) \bmod 7 = 4$

h(x)

X	"UBT"	"CSE"	"TIK"	"TOK"
h(x)	4	4	0	4
0			"TIK"	"TIK"
1				
2				
3				
4	"UBT"	"CSE"	"CSE"	"TOK"
5				
6				

h'(x)

X	"UBT"	"CSE"	"TIK"	"TOK"
h'(x)	3	6	1	4
0				
1				
2				
3		"UBT"	"UBT"	"UBT"
4				
5				
6				"CSE"